

Life Expectancy: Africa vs Europe

1. Load the packages

```
library(tidyverse)
library(patchwork)
library(gapminder)
library(ggplot2)
```

2. Explore the Gapminder dataset

First, we look at the gapminder dataset.

```
View(gapminder)
```

We can also look at how many continents are included in the dataset.

There are five continents: **Asia, Europe, Africa, Americas, and Oceania.**

```
unique(gapminder$continent)
```

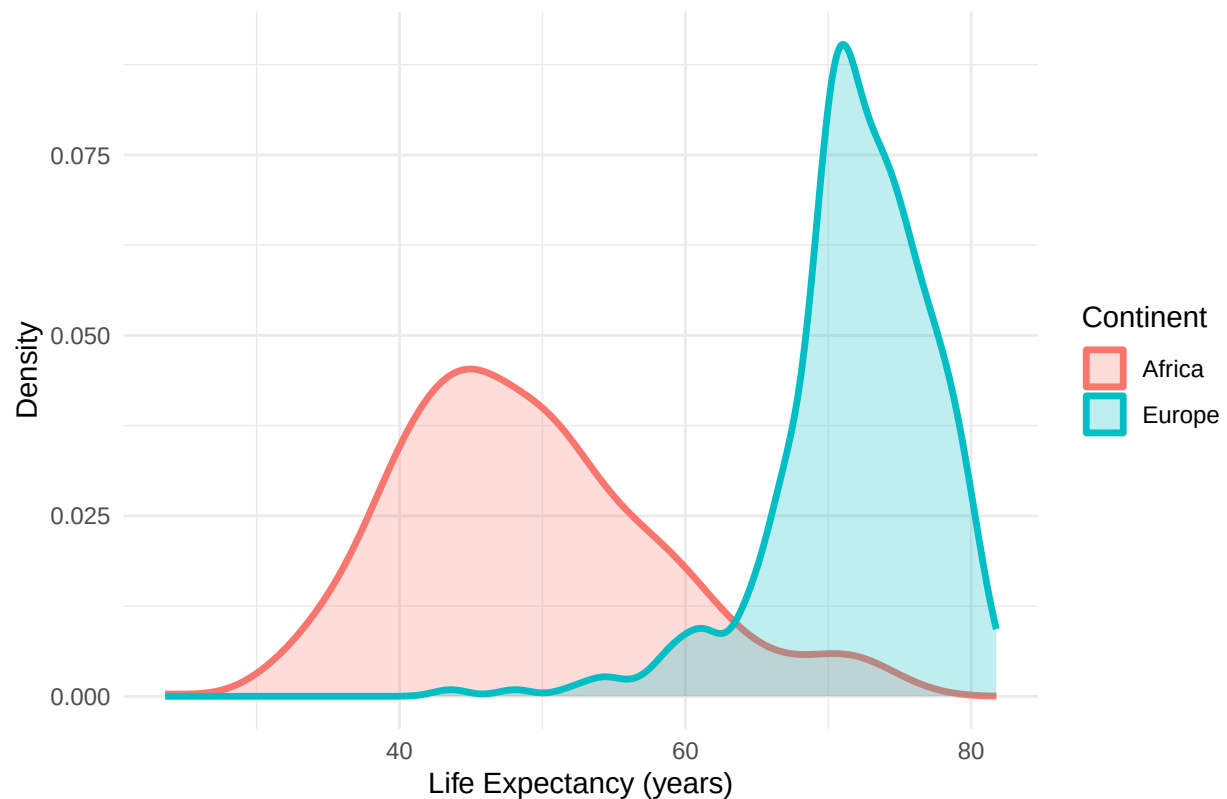
```
## [1] Asia      Europe    Africa    Americas Oceania
## Levels: Africa Americas Asia Europe Oceania
```

3. Compare Life Expectancy between Africa and Europe

Next, we draw a density plot to compare life expectancy between Africa and Europe.

```
ggplot(
  subset(gapminder, continent %in% c("Africa", "Europe")),
  aes(x = lifeExp, fill = continent, color = continent)
) +
  geom_density(alpha = 0.25, linewidth = 1.2) +
  labs(
    title = "Life Expectancy: Africa vs Europe",
    x = "Life Expectancy (years)",
    y = "Density",
    fill = "Continent",
    color = "Continent"
  ) +
  theme_minimal()
```

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4. T-test

Finally, we use a two-sided t-test to compare life expectancy between Africa and Europe.

```
Ttest <- gapminder %>%  
  filter(continent %in% c("Africa", "Europe")) %>%  
  t.test(  
    lifeExp ~ continent,  
    data = .,  
    alternative = "two.sided"  
  )
```

Ttest

```
##  
## Welch Two Sample t-test  
##  
## data: lifeExp by continent  
## t = -49.551, df = 981.2, p-value < 2.2e-16  
## alternative hypothesis: true difference in means between group Africa and group Europe is not equal  
## 95 percent confidence interval:  
## -23.95076 -22.12595  
## sample estimates:  
## mean in group Africa mean in group Europe  
## 48.86533 71.90369
```

The null hypothesis is that the mean life expectancy is the same in Africa and Europe.

The alternative hypothesis is that the mean life expectancy is different between the two continents.

The p-value is less than 0.05, so we reject the null hypothesis. This indicates that there is a statistically significant difference in life expectancy between Africa and Europe.